

A Complete Combinatorial Classification of a Finite Residue-Covering Problem on $[34, 66]$

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Abstract. We present a complete combinatorial classification of a finite residue-covering problem on the interval $I = [34, 66] \cap \mathbb{Z}$. The problem asks for the maximum number of integers in I that can be covered by selecting one residue class modulo 2 and two distinct residue classes modulo each of 3, 5, and 7. While exhaustive search over the 1,260 canonical assignments—equivalently, 10,080 ordered assignments—gives maximum coverage 32, with 126 canonical optima and 1,008 ordered optima, we derive these conclusions combinatorially. By mapping the interval to a punctured 5×7 toroidal grid via the Chinese Remainder Theorem, we reduce the problem to a mod-6 grid-coloring puzzle. We prove the Singleton-Class Theorem, which equates optimal assignments to singleton colors on residual subgrids. By explicitly counting singleton incidences and analyzing color-pair compatibility, we analytically derive the seed distribution $(92, 110, 8)$, proving the exact counts of optimal assignments and the strict maximum coverage bound.

1. Introduction

Finite covering problems occupy a fascinating intersection between elementary number theory and combinatorial geometry. In a typical covering problem, one selects a set of congruence classes and asks what portion of a given domain can be covered. When the domain is a finite interval and the number of selected residue

classes is strictly constrained, the problem often defaults to computational optimization.

In this paper, we investigate a specific finite covering problem on the interval $I = [34, 66]$. The constraints are rigid: one must select exactly one residue class modulo 2, and exactly two distinct residue classes modulo each of 3, 5, and 7. Because the search space is finite, an exhaustive computer search easily reveals the optimal configurations. However, a brute-force algorithm provides the answer without illuminating the underlying structural mechanics.

The objective of this paper is to replace computational enumeration with a formal combinatorial theory. We demonstrate that the optimal covering configurations are not random artifacts of the interval, but are strictly governed by the geometry of a punctured toroidal grid. By reducing the arithmetic congruences to a grid-coloring problem, we provide a purely structural proof of the maximum coverage bound and the exact enumeration of all optimal assignments.

2. Related Work

The study of covering systems of congruences was famously initiated by Paul Erdős in 1950 [1], who asked whether there exist systems of congruences with distinct moduli that cover the integers. The evolution of this problem has generated a vast literature, expertly surveyed by Porubský and Schönheim [4] and cataloged by Guy [2]. While infinite covering systems remain a central topic in combinatorial number theory, finite interval covering with strict multiplicity constraints is often treated via integer linear programming or algorithmic sieve methods, largely due to the general algorithmic hardness of finite set covering problems [5].

Finite-interval variants of congruence covering problems have also been studied. Myerson, Poon, and Simpson introduced incongruent restricted disjoint covering systems on intervals $[1, n]$, requiring each integer to belong to exactly one selected congruence class and each class to occur at least twice within the interval. Their

work combines computational and structural methods and provides a particularly relevant finite-interval precedent [6]. Although the present problem differs by fixing the moduli 2, 3, 5, 7, prescribing residue multiplicities, and maximizing rather than exactly partitioning coverage, both settings seek structural explanations for finite congruence configurations.

Recent work on Erdős covering systems has also developed structural, density, and enumeration methods for unions of arithmetic progressions [7–10]. The present problem is substantially smaller and more rigid: it concerns a single finite interval and a fixed collection of prime moduli. Its contribution is an exact CRT-grid classification of all extremal assignments rather than an asymptotic or global covering theorem.

This paper bridges the gap between algorithmic search and structural combinatorics. By utilizing the Chinese Remainder Theorem (CRT) to map a truncated interval onto a toroidal grid, we draw on techniques analogous to those found in combinatorial design theory and grid-coloring [3]. The reduction of a 10,080-state search space to a localized 20-case compatibility check demonstrates how classical CRT isomorphisms can expose hidden geometric rigidity in seemingly unstructured finite optimization problems.

3. Notation

Symbol	Definition
I	The integer interval $[34, 66] \cap \mathbb{Z}$.
$\mathcal{A}_{\text{can}}$	A canonical assignment (P_2, P_3, P_5, P_7) .
$\mathcal{A}_{\text{ord}}$	An ordered assignment with distinct residue pairs.
$\mathcal{G}$	The punctured 5×7 toroidal CRT grid.
$\phi(n)$	The cell correspondence map $\phi(n) = (n \bmod 5, n \bmod 7)$.
$c(x, y)$	The color of a cell, defined as $n \bmod 6$.
B	A seed (P_5, P_7) representing deleted rows and columns.
$R(B)$	The residual board of surviving cells for seed B .
$s(B)$	The number of singleton colors on $R(B)$.
$\psi(P_2, P_3)$	The bijection mapping a completion to a mod-6 color.
$\rho_j(c)$	The row occupied by color j in column c .
$\mathcal{R}_j(D)$	The family of valid deleted row pairs making color j a singleton for deleted column pair D .
Q_{ij}	The number of seeds making colors i and j simultaneous singletons.
N_k	The number of seeds producing exactly k singleton colors.

4. Definitions and CRT Model

Definition 4.1 (Ordered and Canonical Assignments). An **ordered assignment** is a tuple $\mathcal{A}_{\text{ord}} = (r_2, (a_3, b_3), (a_5, b_5), (a_7, b_7))$ where $r_2 \in \mathbb{Z}/2\mathbb{Z}$ and $(a_p, b_p) \in (\mathbb{Z}/p\mathbb{Z})^2$ are **distinct** ordered pairs (i.e., $a_p \neq b_p$) for $p \in \{3, 5, 7\}$. A **canonical assignment** is a tuple $\mathcal{A}_{\text{can}} = (P_2, P_3, P_5, P_7)$ where $P_2 \subset \mathbb{Z}/2\mathbb{Z}$ with $|P_2| = 1$, and $P_p \subset \mathbb{Z}/p\mathbb{Z}$ are unordered sets with $|P_p| = 2$ for $p \in \{3, 5, 7\}$.

Definition 4.2 (Coverage and Optimum). For a canonical assignment $\mathcal{A}_{\text{can}}$, an integer $n \in I$ is **covered** if $n \bmod p \in P_p$ for at least one $p \in \{2, 3, 5, 7\}$. The **coverage** of $\mathcal{A}_{\text{can}}$, denoted $\text{cov}(\mathcal{A}_{\text{can}})$, is the cardinality of the set of covered integers in I . An assignment is **optimal** if its coverage achieves the maximum possible value over all canonical assignments.

Definition 4.3 (The CRT Grid and Cell Correspondence). Because $\gcd(5, 7) = 1$, the Chinese Remainder Theorem (CRT) provides a natural bijection between the residue classes modulo 35 and the coordinates of a 5×7 toroidal grid, $\mathbb{Z}/5\mathbb{Z} \times \mathbb{Z}/7\mathbb{Z}$. We define the **cell correspondence** as the injective map $\phi : I \rightarrow \mathbb{Z}/5\mathbb{Z} \times \mathbb{Z}/7\mathbb{Z}$ given by:

$$\phi(n) = (n \bmod 5, n \bmod 7)$$

We identify each integer $n \in I$ with its corresponding cell $(x, y) = \phi(n)$.

Definition 4.4 (The Punctures). The interval I spans 33 consecutive integers. Modulo 35, the elements of I represent every residue class except 32 and 33. We define the **punctures** as the grid coordinates corresponding to these two missing residue classes:

- $32 \mapsto (32 \bmod 5, 32 \bmod 7) = (2, 4)$
- $33 \mapsto (33 \bmod 5, 33 \bmod 7) = (3, 5)$

The **punctured grid** $\mathcal{G}$ is defined as the image of I under ϕ :

$$\mathcal{G} = (\mathbb{Z}/5\mathbb{Z} \times \mathbb{Z}/7\mathbb{Z}) \setminus \{(2, 4), (3, 5)\}$$

Thus, $\mathcal{G}$ contains exactly 33 cells, establishing a perfect bijection between I and $\mathcal{G}$.

Definition 4.5 (The Mod-6 Colors). We define a coloring function $c : \mathcal{G} \rightarrow \mathbb{Z}/6\mathbb{Z}$ that assigns to each cell $(x, y) = \phi(n)$ the **color** $c(x, y) = n \bmod 6$.

Definition 4.6 (Seed, Deleted Rows, and Deleted Columns). A **seed** B is the projection of a canonical assignment onto its modulo-5 and modulo-7 components: $B = \pi(\mathcal{A}_{\text{can}}) = (P_5, P_7)$. Geometrically on $\mathcal{G}$, the two elements of P_5 specify two **deleted rows**, and the two elements of P_7 specify two **deleted columns**. There are exactly $\binom{5}{2} \binom{7}{2} = 210$ possible seeds.

Definition 4.7 (Residual Board). For a given seed $B = (P_5, P_7)$, the **residual board** $R(B)$ is the set of surviving cells in $\mathcal{G}$ that do not lie in the deleted rows or columns:

$$R(B) = \{(x, y) \in \mathcal{G} \mid x \notin P_5 \text{ and } y \notin P_7\}$$

Definition 4.8 (Singleton Color). For a fixed seed B , a color $j \in \mathbb{Z}/6\mathbb{Z}$ is a **singleton color** if exactly one cell on the residual board $R(B)$ has color j . We denote the total number of singleton colors on the residual board of seed B as $s(B)$.

5. Structural Reductions

Lemma 5.1 (Residual Equivalence). Let $\mathcal{A}_{\text{can}} = (P_2, P_3, P_5, P_7)$ be a canonical assignment, and let $B = (P_5, P_7)$ be its associated seed. An integer $n \in I$ is not covered by $P_5 \cup P_7$ if and only if its corresponding cell $\phi(n)$ lies on the residual board $R(B)$.

Proof. By Definition 4.3 and 4.7, $\phi(n) \in R(B) \iff n \bmod 5 \notin P_5 \text{ and } n \bmod 7 \notin P_7$. This is a strict set-theoretic equivalence. $\square$

Lemma 5.2 (Mod-6 Completion Bijection). Let $\mathcal{C}$ be the set of all possible canonical completions (P_2, P_3) . There exists a strict bijection $\psi : \mathcal{C} \rightarrow \mathbb{Z}/6\mathbb{Z}$ such that an integer n is not covered by $P_2 \cup P_3$ if and only if $n \equiv \psi(P_2, P_3) \pmod{6}$.

Proof. A canonical completion omits exactly one parity class $p_2 \in \mathbb{Z}/2\mathbb{Z}$ and one residue class $p_3 \in \mathbb{Z}/3\mathbb{Z}$. An integer n survives $P_2 \cup P_3$ if and only if $n \equiv p_2 \pmod{2}$ and $n \equiv p_3 \pmod{3}$. Because $\gcd(2, 3) = 1$, the Chinese Remainder Theorem guarantees a unique solution $j \in \mathbb{Z}/6\mathbb{Z}$. We define $\psi(P_2, P_3) = j$. Since there are exactly $\binom{2}{1} \times \binom{3}{1} = 6$ canonical completions and 6 residue classes modulo 6, ψ is a bijection. $\square$

Theorem 5.3 (Colored-Grid Equivalence and Occupancy Sufficiency). For any canonical assignment $\mathcal{A}_{\text{can}}$, let $B = (P_5, P_7)$ and $j = \psi(P_2, P_3)$. The set of completely uncovered integers (survivors) in I is in exact bijection with the set of cells on the residual board $R(B)$ that have color j . Consequently, the coverage of $\mathcal{A}_{\text{can}}$ is strictly determined by the color occupancy counts of $R(B)$.

Proof. Let $S(\mathcal{A}_{\text{can}})$ be the set of survivors. An integer $n \in S(\mathcal{A}_{\text{can}})$ if and only if it survives $P_5 \cup P_7$ and $P_2 \cup P_3$. By Lemma 5.1, n survives $P_5 \cup P_7 \iff \phi(n) \in R(B)$. By Lemma 5.2, n survives $P_2 \cup P_3 \iff c(\phi(n)) = j$. Because ϕ is a bijection, $|S(\mathcal{A}_{\text{can}})| = |\{(x, y) \in R(B) \mid c(x, y) = j\}|$. The coverage is exactly $33 - |S(\mathcal{A}_{\text{can}})|$, proving occupancy sufficiency. $\square$

6. Singleton-Class Theorem

Definition 6.1 (1-Survivor Assignment). A canonical assignment $\mathcal{A}_{\text{can}}$ is a **1-survivor assignment** if its survivor set has cardinality $|S(\mathcal{A}_{\text{can}})| = 1$.

Theorem 6.2 (Singleton-Class Theorem). For any seed B , the number of canonical completions (P_2, P_3) that extend B into a 1-survivor assignment is exactly $s(B)$, the number of singleton colors on the residual board $R(B)$.

Proof. By Theorem 5.3, for any completion (P_2, P_3) , the number of survivors is $|\{(x, y) \in R(B) \mid c(x, y) = j\}|$, where $j = \psi(P_2, P_3)$. The assignment leaves exactly 1 survivor if and only if color j appears exactly once on $R(B)$; that is, j is a singleton

color. By Lemma 5.2, ψ is a strict bijection. Therefore, every singleton color on $R(B)$ corresponds to exactly one canonical completion yielding a 1-survivor assignment. $\square$

Theorem 6.3 (Uniqueness of Counting). Every 1-survivor canonical assignment is counted exactly once by the sum $\sum_B s(B)$ over all 210 seeds.

Proof. The projection $\pi(\mathcal{A}_{\text{can}}) = (P_5, P_7)$ maps any canonical assignment to a unique seed B . Two distinct canonical assignments cannot project to the same seed and the same completion. Thus, the sets of 1-survivor completions for distinct seeds are disjoint, guaranteeing no assignment is omitted or counted twice. $\square$

7. Singleton Incidence Count

Lemma 7.1 (Color Distribution on $\mathcal{G}$). The 33 cells of $\mathcal{G}$ partition into three 5-cell colors (1, 2, 3) and three 6-cell colors (0, 4, 5).

Proof. The interval $I = [34, 66]$ contains 33 integers. Because $33 = 5 \times 6 + 3$, exactly three residue classes modulo 6 appear 6 times, and three appear 5 times. The first three integers are $34 \equiv 4$, $35 \equiv 5$, and $36 \equiv 0 \pmod{6}$. Thus, colors 0, 4, 5 have 6 cells, and 1, 2, 3 have 5 cells. $\square$

Lemma 7.2 (Grid Geometry of Colors). For any color j , no two of its cells share the same column. A 5-cell color occupies 5 distinct rows. A 6-cell color occupies all 5 rows, with exactly one row containing two cells (the "double row").

Proof. If $n_1 \equiv n_2 \pmod{6}$ and $n_1 \equiv n_2 \pmod{7}$, then $n_2 - n_1$ is a multiple of 42. Since the maximum difference in I is 32, no two cells of the same color share a column. Consecutive integers of a fixed color modulo 6 differ by 6. Their row coordinates therefore increase by $6 \equiv 1 \pmod{5}$. For a five-cell color, the five row coordinates are consequently five consecutive residues modulo 5, hence are all distinct. For a six-cell

color, the six row coordinates consist of all five residues, with the initial residue repeated exactly once. This proves the double-row assertion directly. $\square$

Lemma 7.3 (5-Cell Singleton Count). Each 5-cell color becomes a singleton on exactly 30 residual boards.

Proof. To reduce a 5-cell color to a singleton, a seed must delete exactly 4 of its cells. We choose 2 columns from the 5 it occupies ($\binom{5}{2} = 10$ ways), deleting 2 cells. The remaining 3 cells occupy 3 distinct rows. We choose 2 rows from these 3 ($\binom{3}{2} = 3$ ways) to delete the remaining 2 cells. Total seeds: $10 \times 3 = 30$. $\square$

Lemma 7.4 (6-Cell Singleton Count). Each 6-cell color becomes a singleton on exactly 12 residual boards.

Proof. To remove 5 cells from a 6-cell color, the deleted row pair P_5 must include the double row. If the double row were not deleted, then the two deleted rows could remove at most two cells, and the two deleted columns could remove at most two additional cells because the color occupies distinct columns. At least two cells would remain. Thus, the double row must be one of the deleted rows (removing 2 cells). P_5 must also include exactly 1 additional row from the remaining 4 rows ($\binom{4}{1} = 4$ ways). The remaining 3 cells occupy 3 distinct columns. P_7 must be chosen from these 3 columns ($\binom{3}{2} = 3$ ways). Total seeds: $1 \times 4 \times 3 = 12$. $\square$

Theorem 7.5 (Singleton Incidence Theorem). Across all 210 seeds, the total number of singleton colors is exactly 126. Formally: $\sum_B s(B) = 126$.

Proof. By Lemma 7.1, there are three 5-cell colors and three 6-cell colors. By Lemmas 7.3 and 7.4, the total is $3(30) + 3(12) = 90 + 36 = 126$. $\square$

8. Pair Compatibility

Definition 8.1 (Row-Pair Family). For every color $j \in \mathbb{Z}/6\mathbb{Z}$ and deleted column pair $D \in \binom{\mathbb{Z}/7\mathbb{Z}}{2}$, we define the family of valid deleted row pairs:

$$\mathcal{R}_j(D) = \left\{ R \in \binom{\mathbb{Z}/5\mathbb{Z}}{2} \mid \text{deleting } R \text{ and } D \text{ leaves exactly one cell of color } j \right\}$$

The number of seeds making both colors i and j singletons simultaneously is $Q_{ij} = \sum_D |\mathcal{R}_i(D) \cap \mathcal{R}_j(D)|$.

Lemma 8.2 (6-Cell Incompatibility). If at least one of the colors i, j is a 6-cell color, then $Q_{ij} = 0$.

Proof. The proof replaces exhaustive enumeration of the 1,260 canonical assignments or 210 seeds with a structural reduction to 20 local compatibility cases. Let a be a 6-cell color with double row r_a^* . For a to become a singleton, the deleted row pair R must contain the double row r_a^* . Thus, $\mathcal{R}_a(D) = \{\{r_a^*, u\}, \{r_a^*, v\}\}$. Any other color b aiming for simultaneous singleton status must tolerate the deletion of r_a^* . This strict structural requirement severely limits candidate column pairs D . A complete finite compatibility classification (Appendix A) exhaustively evaluates these 20 candidate column pairs for all 12 color pairs involving $\{0, 4, 5\}$. In every case, $\mathcal{R}_i(D) \cap \mathcal{R}_j(D) = \emptyset$. Thus, $Q_{ij} = 0$. $\square$

Lemma 8.3 (5-Cell Compatibility). For the 5-cell colors $\{1, 2, 3\}$, the simultaneous singleton counts are $Q_{12} = 3$, $Q_{23} = 3$, and $Q_{13} = 2$.

Proof. Let $\rho_j(c)$ denote the row occupied by color j in column c . For $\mathcal{R}_j(D)$ to be non-empty, D must be a subset of occupied columns C_j . The remaining 3 cells occupy the row set $T_j(D) = \mathbb{Z}/5\mathbb{Z} \setminus \rho_j(D)$. To leave exactly one cell, $\mathcal{R}_j(D) = \binom{T_j(D)}{2}$. For two 5-cell colors i, j , the number of compatible row pairs is $\binom{|T_i(D) \cap T_j(D)|}{2}$.

- Q_{12} : $C_1 \cap C_2 = \{0, 1, 2, 6\}$. Exactly three column pairs yield $|T_1 \cap T_2| = 2$: $D \in \{\{0, 2\}, \{1, 6\}, \{2, 6\}\}$. Each contributes $\binom{2}{2} = 1$ valid row pair. Thus,

$$Q_{12} = 3.$$

- Q_{23} : $C_2 \cap C_3 = \{0, 1, 2, 3\}$. Exactly three column pairs yield $|T_2 \cap T_3| = 2$: $D \in \{\{0, 2\}, \{0, 3\}, \{1, 3\}\}$. Thus, $Q_{23} = 3$.
- Q_{13} : $C_1 \cap C_3 = \{0, 1, 2\}$. Exactly two column pairs yield $|T_1 \cap T_3| = 2$: $D \in \{\{0, 1\}, \{1, 2\}\}$. Thus, $Q_{13} = 2$.

Lemma 8.4 (Impossibility of Triple Singletons). No seed produces three singleton colors ($Q_{123} = 0$). $\square$

Proof. By Lemma 8.2, any triple must consist of $\{1, 2, 3\}$. Each 5-cell color misses exactly two columns: $M_1 = \{3, 4\}$, $M_2 = \{4, 5\}$, $M_3 = \{5, 6\}$. For D to be valid for all three, it must avoid $M_1 \cup M_2 \cup M_3 = \{3, 4, 5, 6\}$. The complement is $\{0, 1, 2\}$, so $D \in \{\{0, 1\}, \{0, 2\}, \{1, 2\}\}$. Evaluating the triple intersection $T_1(D) \cap T_2(D) \cap T_3(D)$ yields $\emptyset$, $\{0\}$, and $\emptyset$ respectively. Because the intersection size is never ≥ 2 , $\binom{\leq 1}{2} = 0$. $\square$

Theorem 8.5 (Total Pair Count). For any seed B , $s(B) \leq 2$, and the total number of pairs of singletons across all seeds is exactly 8. Formally: $\sum_B \binom{s(B)}{2} = 8$.

Proof. By Lemma 8.4, $s(B) \leq 2$. Each unordered pair of singleton colors on a seed contributes exactly once to exactly one Q_{ij} . By Lemma 8.2, all terms involving $\{0, 4, 5\}$ are zero. By Lemma 8.3, the sum is $Q_{12} + Q_{23} + Q_{13} = 3 + 3 + 2 = 8$. $\square$

9. Distribution Theorem

Theorem 9.1 (Seed Distribution). The 210 seeds partition exactly into 92 seeds with no singleton color, 110 seeds with one singleton color, and 8 seeds with two singleton colors.

Proof. Let N_k denote the number of seeds B for which $s(B) = k$. By Theorem 8.5, $N_k = 0$ for $k \geq 3$. We construct a system of three equations:

$$1. \text{ Total Seeds: } N_0 + N_1 + N_2 = \binom{5}{2} \binom{7}{2} = 210$$

2. Total Singletons (Theorem 7.5): $N_1 + 2N_2 = 126$

3. Total Pairs (Theorem 8.5): $N_2 = 8$

Substituting (3) into (2) yields $N_1 = 126 - 16 = 110$. Substituting N_1, N_2 into (1) yields $N_0 = 210 - 118 = 92$. Thus, $(N_0, N_1, N_2) = (92, 110, 8)$. $\square$

10. Enumeration Consequences

Theorem 10.1 (Canonical Optima Count). *There are exactly 126 canonical 1-survivor assignments.*

Proof. By Theorem 6.2, the number of 1-survivor canonical completions for a seed B is exactly $s(B)$. By Theorem 6.3, summing $s(B)$ counts every 1-survivor canonical assignment exactly once. By Theorem 9.1, the total is $\sum s(B) = 0(92) + 1(110) + 2(8) = 126$. $\square$

Theorem 10.2 (Ordered Optima Count). *There are exactly 1,008 ordered 1-survivor assignments.*

Proof. By Definition 4.1, every ordered assignment requires distinct residues at the odd primes ($a_p \neq b_p$). Therefore, for any canonical assignment, each 2-element set P_p has exactly $2! = 2$ valid orderings. Because the choices of ordering at $p = 3, 5, 7$ are independent, every canonical assignment corresponds to exactly $2^3 = 8$ distinct ordered assignments. By Theorem 10.1, there are 126 canonical 1-survivor assignments. Thus, the number of ordered 1-survivor assignments is $126 \times 8 = 1008$. $\square$

11. Maximum-Coverage Theorem

Theorem 11.1 (Maximum Coverage). *For the interval $I = [34, 66]$, no assignment covers all 33 integers. The maximum attainable coverage is exactly 32.*

Proof. To achieve a coverage of 33, an assignment must leave 0 survivors. By Theorem 5.3, 0 survivors requires that color $j = \psi(P_2, P_3)$ has exactly 0 cells on the residual board $R(B)$.

By Lemma 7.1, every color has 5 or 6 cells on $\mathcal{G}$. By Lemma 7.2, no two cells of the same color share a column.

For a 5-cell color, deleting 2 rows and 2 columns removes a maximum of $2 + 2 = 4$ cells, leaving ≥ 1 cell on $R(B)$.

For a 6-cell color, if the double row is deleted, two rows remove at most three cells and two columns remove at most two more; if it is not deleted, two rows remove at most two cells and two columns remove at most two more. In all cases, at least $6 - 5 = 1$ cell survives.

Because every color has at least 1 surviving cell on every possible residual board, no completion can leave 0 survivors. Thus, the maximum possible coverage is ≤ 32 .

By Theorem 10.1, there are exactly 126 canonical assignments that leave exactly 1 survivor. Because $126 > 0$, a coverage of 32 is strictly attainable. Therefore, the maximum coverage is exactly 32. $\square$

12. Discussion and Conclusion

This paper provides a complete combinatorial classification of the residue-covering problem on [34, 66]. By mapping the arithmetic congruences to a punctured 5×7 toroidal grid, we demonstrated that the problem's constraints are fundamentally geometric. The reduction sequence—from assignments to residual boards, to mod-6 colors, to singleton counts—preserves the exact multiplicity of the solution space while stripping away irrelevant arithmetic data.

The most significant result of this framework is the replacement of a 10,080-state exhaustive computer search with a structural derivation. The distribution of seeds $(92, 110, 8)$ is not an empirical observation but a consequence of the row-column

incidence structure of the punctured CRT grid: specifically, the interaction between the 6 diagonal color lines and the 2 missing punctures. The reduction of the pair-compatibility problem to 20 localized cases (Appendix A) shows that the compatibility analysis forces exactly eight double-singleton seeds.

The combinatorial theory provides an independent derivation of the classification. Exhaustive computation serves as an external verification of the resulting counts. It explains not only *what* the optimal assignments are, but *why* the geometry of the interval strictly bounds the maximum coverage at 32 and forces exactly 1,008 ordered optima.

13. References

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Appendix A: Row Maps and Compatibility Classification

A.1 Explicit Row Maps $\rho_j(c)$

For each color $j \in \mathbb{Z}/6\mathbb{Z}$, $\rho_j(c)$ is the row occupied by color j in column c , derived directly from $(n \bmod 5, n \bmod 7)$ for $n \in [34, 66]$:

- **6-cell colors (0, 4, 5):**
 - $\rho_0: 0 \mapsto 2, 1 \mapsto 1, 3 \mapsto 1, 4 \mapsto 0, 5 \mapsto 4, 6 \mapsto 3$. (Double row: 1 at cols 1, 3)
 - $\rho_4: 1 \mapsto 4, 2 \mapsto 3, 3 \mapsto 2, 4 \mapsto 1, 5 \mapsto 0, 6 \mapsto 4$. (Double row: 4 at cols 1, 6)
 - $\rho_5: 0 \mapsto 0, 2 \mapsto 0, 3 \mapsto 4, 4 \mapsto 3, 5 \mapsto 2, 6 \mapsto 1$. (Double row: 0 at cols 0, 2)
- **5-cell colors (1, 2, 3):**
 - $\rho_1: 0 \mapsto 4, 1 \mapsto 3, 2 \mapsto 2, 5 \mapsto 1, 6 \mapsto 0$.
 - $\rho_2: 0 \mapsto 1, 1 \mapsto 0, 2 \mapsto 4, 3 \mapsto 3, 6 \mapsto 2$.
 - $\rho_3: 0 \mapsto 3, 1 \mapsto 2, 2 \mapsto 1, 3 \mapsto 0, 4 \mapsto 4$.

A.2 Exhaustive Candidate Generation

For a 6-cell color a to become a singleton, D must hit exactly two of its cells, and neither can be in the double row. Thus, D must be a subset of the valid columns V_a :

- $V_0 = \{0, 4, 5, 6\}$, $V_4 = \{2, 3, 4, 5\}$, $V_5 = \{3, 4, 5, 6\}$

For a 5-cell color b , D must be a subset of its occupied columns V_b :

- $V_1 = \{0, 1, 2, 5, 6\}$, $V_2 = \{0, 1, 2, 3, 6\}$, $V_3 = \{0, 1, 2, 3, 4\}$

For any pair $\{i, j\}$, the exhaustive candidate column pairs D for which both $\mathcal{R}_i(D)$ and $\mathcal{R}_j(D)$ can be non-empty are strictly the 2-element subsets of $V_i \cap V_j$.

A.3 Finite Compatibility Table

For a 6-cell color a with double row r_a^* , $\mathcal{R}_a(D) = \{\{r_a^*, u\}, \{r_a^*, v\}\}$, where u, v are the rows of the remaining cells. For a 5-cell color b , $\mathcal{R}_b(D)$ is the set of all pairs from the 3 remaining rows. The table below evaluates every valid candidate $D \in \binom{V_i \cap V_j}{2}$. In every case, the intersection of valid row pairs is strictly empty.

$\{i, j\}$	D	$\mathcal{R}_i(D)$	$\mathcal{R}_j(D)$	$\mathcal{R}_i(D) \cap \mathcal{R}_j(D)$
$\{0, 1\}$	$\{0, 5\}$	$\{\{1, 0\}, \{1, 3\}\}$	$\{\{0, 2\}, \{0, 3\}, \{2, 3\}\}$	$\emptyset$
	$\{0, 6\}$	$\{\{1, 0\}, \{1, 4\}\}$	$\{\{1, 2\}, \{1, 3\}, \{2, 3\}\}$	$\emptyset$
	$\{5, 6\}$	$\{\{1, 2\}, \{1, 0\}\}$	$\{\{2, 3\}, \{2, 4\}, \{3, 4\}\}$	$\emptyset$
$\{0, 2\}$	$\{0, 6\}$	$\{\{1, 0\}, \{1, 4\}\}$	$\{\{0, 3\}, \{0, 4\}, \{3, 4\}\}$	$\emptyset$
$\{0, 3\}$	$\{0, 4\}$	$\{\{1, 4\}, \{1, 3\}\}$	$\{\{0, 1\}, \{0, 2\}, \{1, 2\}\}$	$\emptyset$
$\{0, 4\}$	$\{4, 5\}$	$\{\{1, 2\}, \{1, 3\}\}$	$\{\{4, 3\}, \{4, 2\}\}$	$\emptyset$
$\{0, 5\}$	$\{4, 5\}$	$\{\{1, 2\}, \{1, 3\}\}$	$\{\{0, 4\}, \{0, 1\}\}$	$\emptyset$
	$\{4, 6\}$	$\{\{1, 2\}, \{1, 4\}\}$	$\{\{0, 4\}, \{0, 2\}\}$	$\emptyset$
	$\{5, 6\}$	$\{\{1, 2\}, \{1, 0\}\}$	$\{\{0, 4\}, \{0, 3\}\}$	$\emptyset$
$\{1, 4\}$	$\{2, 5\}$	$\{\{0, 3\}, \{0, 4\}, \{3, 4\}\}$	$\{\{4, 1\}, \{4, 2\}\}$	$\emptyset$
$\{1, 5\}$	$\{5, 6\}$	$\{\{2, 3\}, \{2, 4\}, \{3, 4\}\}$	$\{\{0, 4\}, \{0, 3\}\}$	$\emptyset$
$\{2, 4\}$	$\{2, 3\}$	$\{\{0, 1\}, \{0, 2\}, \{1, 2\}\}$	$\{\{4, 1\}, \{4, 0\}\}$	$\emptyset$
$\{2, 5\}$	$\{3, 6\}$	$\{\{0, 1\}, \{0, 4\}, \{1, 4\}\}$	$\{\{0, 3\}, \{0, 2\}\}$	$\emptyset$
$\{3, 4\}$	$\{2, 3\}$	$\{\{2, 3\}, \{2, 4\}, \{3, 4\}\}$	$\{\{4, 1\}, \{4, 0\}\}$	$\emptyset$
	$\{2, 4\}$	$\{\{0, 2\}, \{0, 3\}, \{2, 3\}\}$	$\{\{4, 2\}, \{4, 0\}\}$	$\emptyset$
	$\{3, 4\}$	$\{\{1, 2\}, \{1, 3\}, \{2, 3\}\}$	$\{\{4, 3\}, \{4, 0\}\}$	$\emptyset$
$\{3, 5\}$	$\{3, 4\}$	$\{\{1, 2\}, \{1, 3\}, \{2, 3\}\}$	$\{\{0, 2\}, \{0, 1\}\}$	$\emptyset$
$\{4, 5\}$	$\{3, 4\}$	$\{\{4, 3\}, \{4, 0\}\}$	$\{\{0, 2\}, \{0, 1\}\}$	$\emptyset$

$\{i, j\}$	D	$\mathcal{R}_i(D)$	$\mathcal{R}_j(D)$	$\mathcal{R}_i(D) \cap \mathcal{R}_j(D)$
	$\{3, 5\}$	$\{\{4, 3\}, \{4, 1\}\}$	$\{\{0, 3\}, \{0, 1\}\}$	$\emptyset$
	$\{4, 5\}$	$\{\{4, 3\}, \{4, 2\}\}$	$\{\{0, 4\}, \{0, 1\}\}$	$\emptyset$